Title: Agriculture AI for smart farming

Abstract

Agriculture plays a vital role in ensuring food security and sustainable development. Traditional farming methods often rely on manual monitoring and experience, making it difficult to accurately predict crop diseases, irrigation requirements, and yield. This project proposes an **Agriculture AI for Smart Farming** framework that utilizes Machine Learning techniques to support farmers in making informed decisions and improving agricultural productivity.

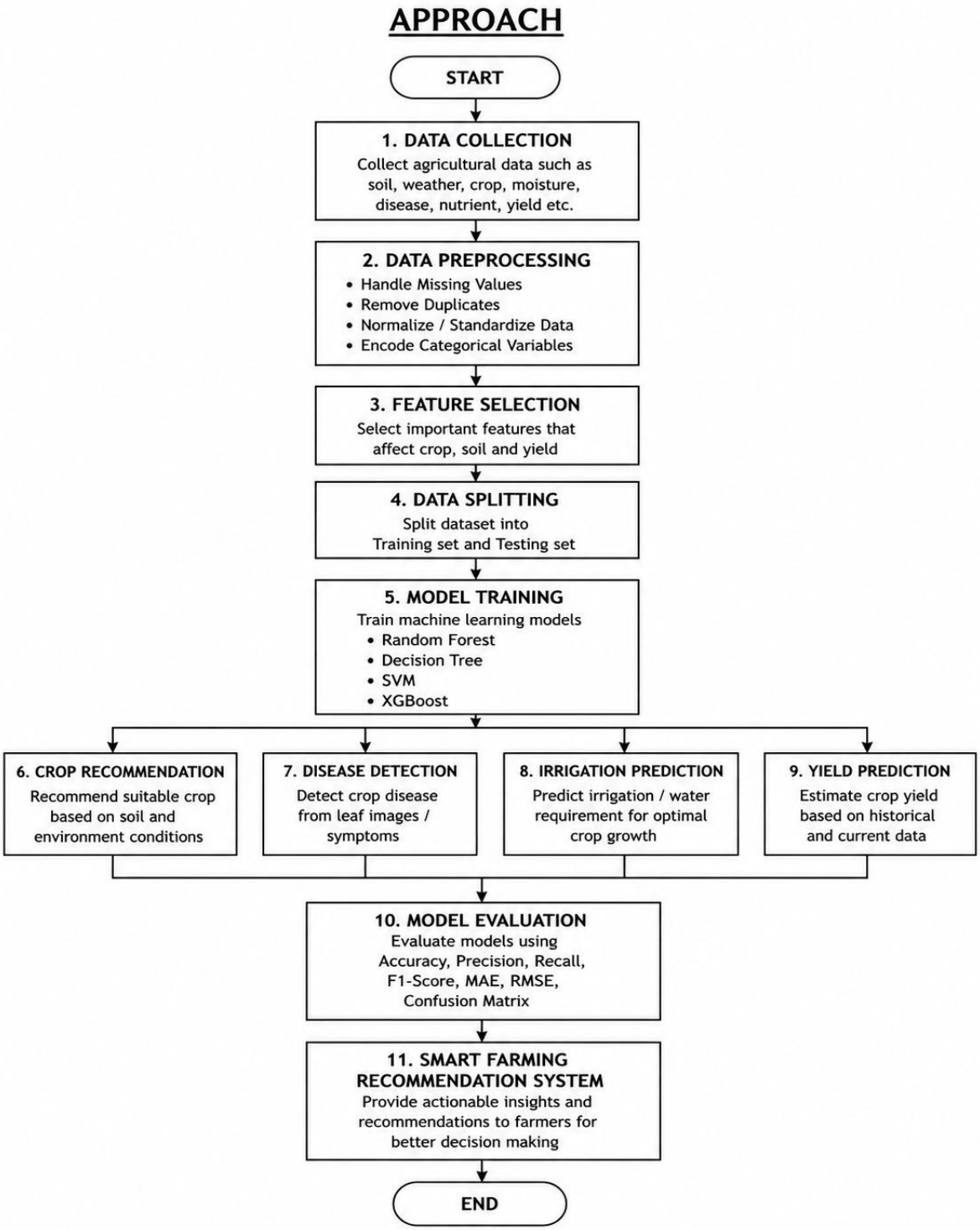
The proposed system collects agricultural data such as soil moisture, temperature, humidity, rainfall, soil nutrients, and crop health information. Data preprocessing techniques are applied to remove missing values and normalize the dataset. Machine Learning algorithms such as **Random Forest, Decision Tree, Support Vector Machine (SVM), and XGBoost** are trained to predict crop diseases, recommend suitable crops, estimate irrigation needs, and forecast crop yield. The framework can also integrate weather information to provide timely farming recommendations.

The performance of the proposed model is evaluated using **Accuracy, Precision, Recall, F1-Score, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Confusion Matrix**, depending on the prediction task. The resulting system aims to improve crop productivity, reduce water wastage, minimize disease outbreaks, and support sustainable farming practices through intelligent and data-driven decision-making.

Datasets

Dataset	URL link
Crop Recommendation Dataset	https://huggingface.co/datasets/THUDM/Crop_recommendation
PlantVillage Classification	https://huggingface.co/datasets/keremberke/plantvillage-classification
PlantVillage	https://huggingface.co/datasets/timm/plantvillage

Approach


APPROACH
START
1. DATA COLLECTION
Collect agricultural data such as soil, weather, crop, moisture, disease, nutrient, yield etc.
2. DATA PREPROCESSING
• Handle Missing Values
• Remove Duplicates
• Normalize / Standardize Data
• Encode Categorical Variables
3. FEATURE SELECTION
Select important features that affect crop, soil and yield
4. DATA SPLITTING
Split dataset into Training set and Testing set
5. MODEL TRAINING
Train machine learning models
• Random Forest
• Decision Tree
• SVM
• XGBoost
6. CROP RECOMMENDATION
Recommend suitable crop based on soil and environment conditions
7. DISEASE DETECTION
Detect crop disease from leaf images / symptoms
8. IRRIGATION PREDICTION
Predict irrigation / water requirement for optimal crop growth
9. YIELD PREDICTION
Estimate crop yield based on historical and current data
10. MODEL EVALUATION
Evaluate models using Accuracy, Precision, Recall, F1-Score, MAE, RMSE, Confusion Matrix
11. SMART FARMING RECOMMENDATION SYSTEM
Provide actionable insights and recommendations to farmers for better decision making
END

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